

GROUP WORK PROJECT # 1

MScFE 622: Stochastic Modeling

GROUP NUMBER: Not provided

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A - all members contributed.

Executive summary

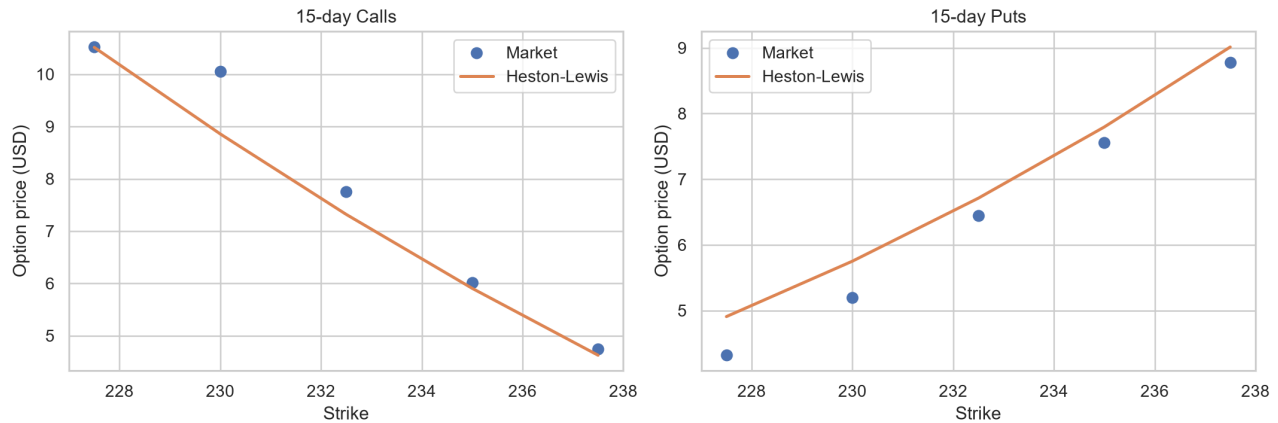
This report presents the assumptions, results, interpretations, and figures for all project questions. SM Energy spot is \$232.90; the annual risk-free rate is 1.50%; one year is 250 trading days; and no dividend yield is assumed.

Step 1 - 15-day derivative

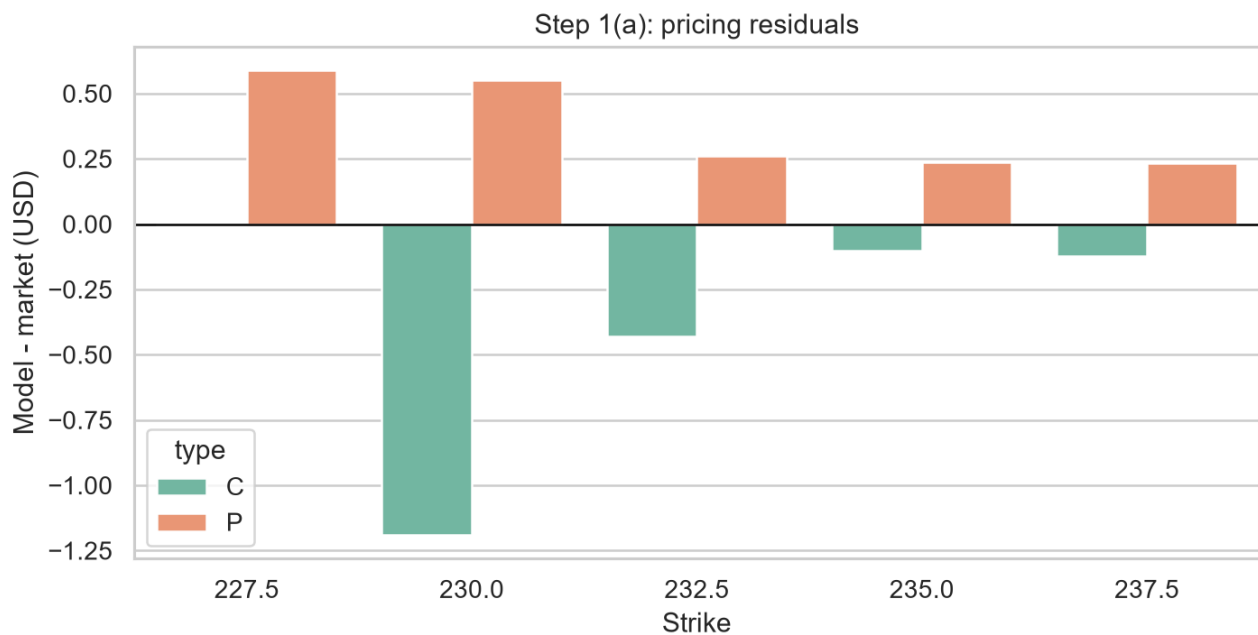
1(a) Heston calibration using Lewis (2001)

The 15-day calls and puts were fitted jointly using the Heston specification and Lewis representation (Heston 327-343; Lewis), with unweighted USD price MSE as required by the project brief (WorldQuant University 1-2). The fitted parameters are $\kappa=2.2032$, $\theta=0.0010$, $\sigma=1.2733$, $\rho=-0.9444$, and initial variance=0.1084. MSE is 0.2458 and RMSE is \$0.4958.

The input quotes have a put-call parity RMSE of \$0.9068; an arbitrage-consistent fit cannot make all residuals zero. Short maturity also weakens identification of long-run variance and mean reversion.



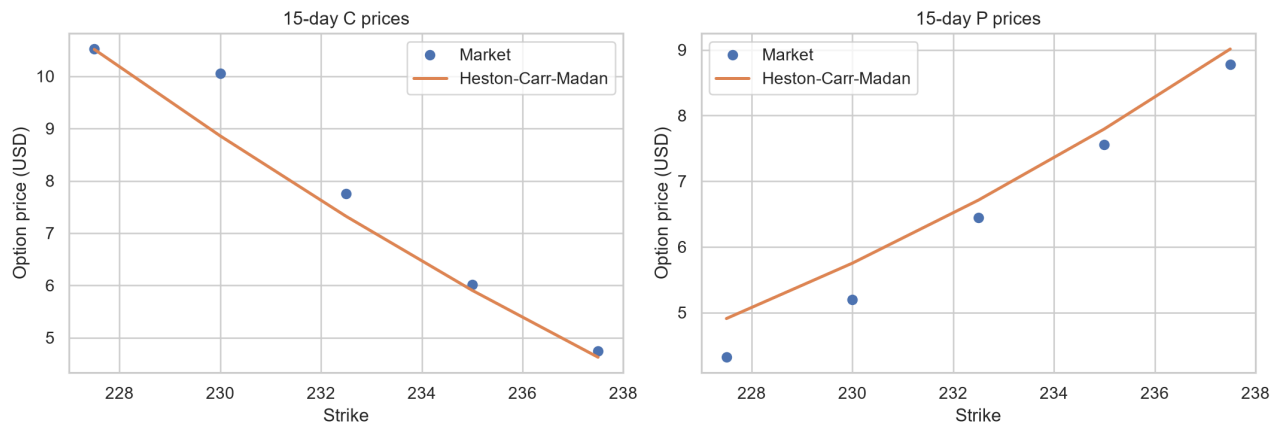
15-day market and model prices



15-day residuals

1(b) Carr-Madan comparison

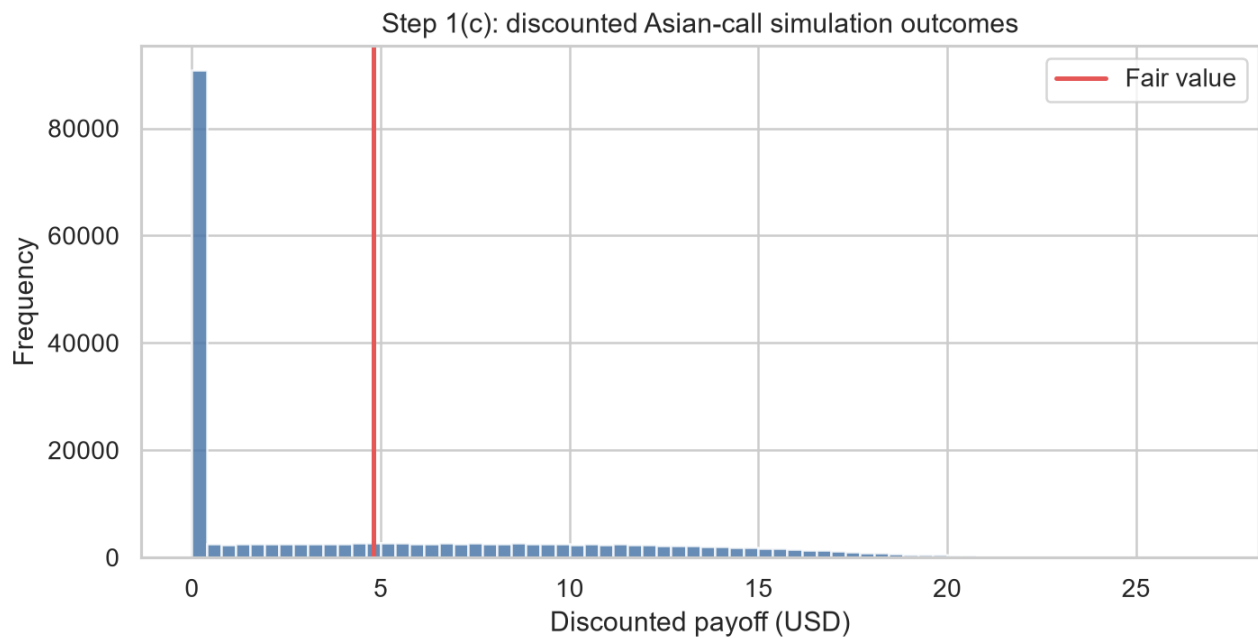
The damped transform follows Carr and Madan (61-73). Its MSE is 0.2458, compared with 0.2458 for Lewis. Fitted prices and numerical stability are the meaningful comparison.



15-day Carr-Madan fit

1(c) 20-day ATM Asian call

Today's spot is included in the arithmetic average, following the project instructions (WorldQuant University 1-2). Across 200,000 risk-neutral scenarios, fair value is \$4.8177 (standard error \$0.0130), with a 95% estimator interval of \$4.7922 to \$4.8432. The 4% fee produces a client price of \$5.0104.

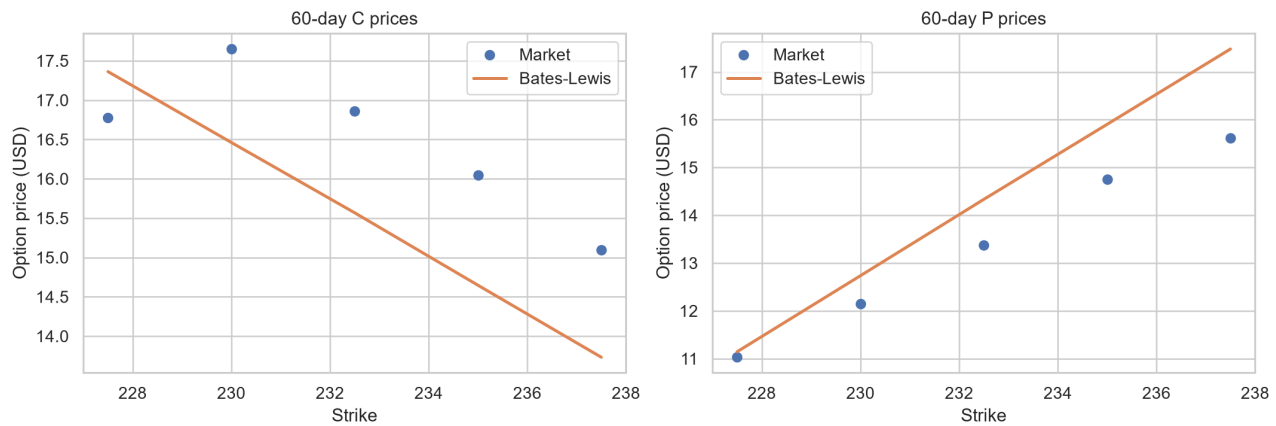


Asian-call outcome distribution

Step 2 - 60-day derivative

2(a) Bates calibration using Lewis

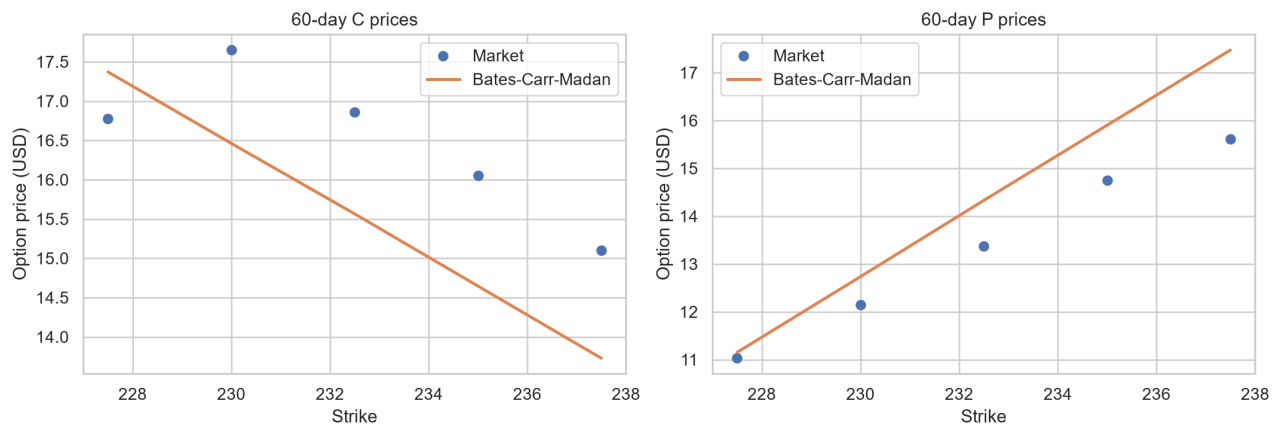
The 60-day jump-diffusion calibration follows Bates (69-107) and has MSE 1.3324 and RMSE \$1.1543.



60-day market and model prices

2(b) Carr-Madan comparison

The matching Carr-Madan calibration follows Carr and Madan (61-73) and has MSE 1.3335. The comparison is based on fitted-price errors and numerical stability.



60-day Carr-Madan fit

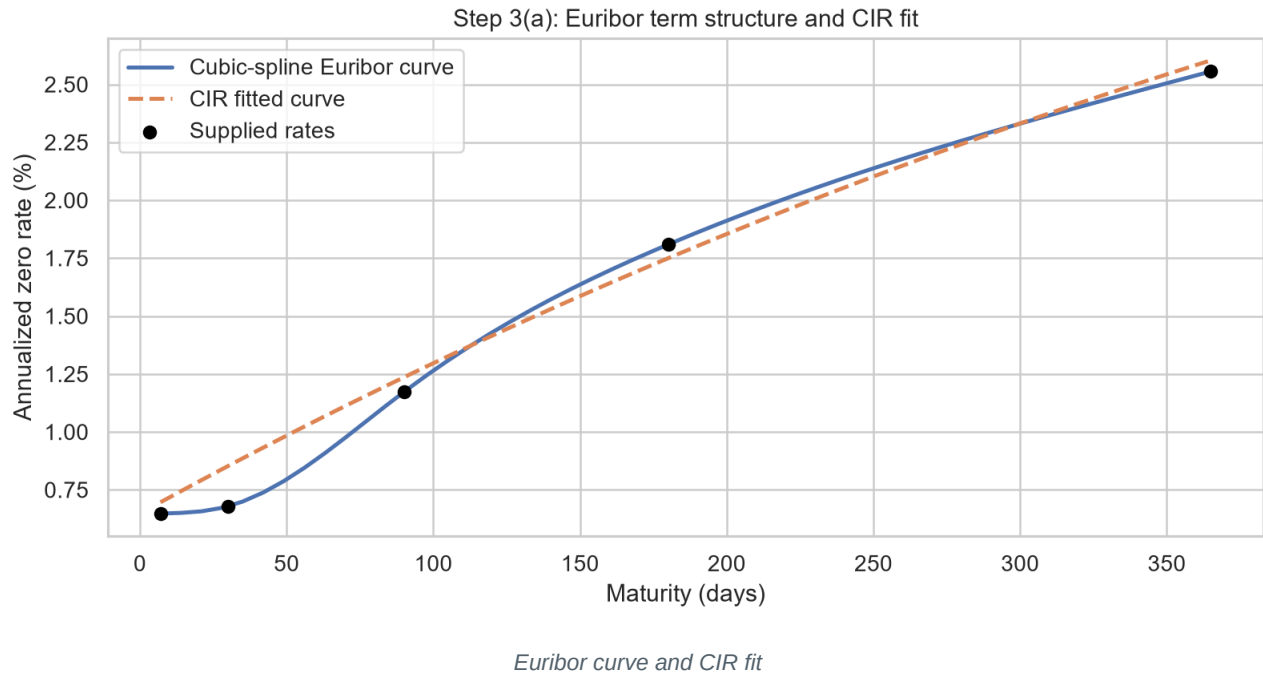
2(c) 70-day 95%-moneyness put

The project specifies a strike of 95% of spot (WorldQuant University 3), giving \$221.25. Fair value is \$8.8187 and the price after the 4% fee is \$9.1715.

Step 3 - Euribor rates

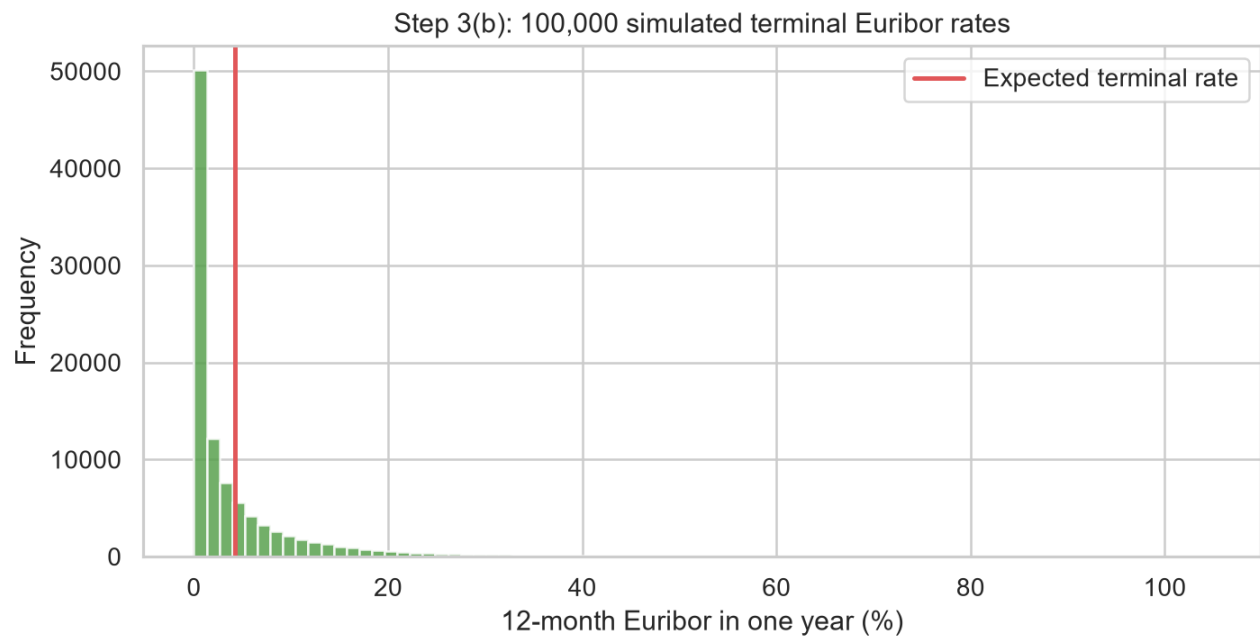
3(a) CIR calibration

The supplied curve contains 0.648% (1 week), 0.679% (1 month), 1.173% (3 months), 1.809% (6 months), and 2.556% (12 months). Weekly interpolation follows the project brief (WorldQuant University 3-4), and the rate model follows Cox, Ingersoll, and Ross (385-407). The fitted values are $a=0.7525$, $b=0.0742$, $\eta=0.5638$; MSE is 0.00000062.



3(b) One-year rate scenarios

Following the required simulation size (WorldQuant University 4), 100,000 daily scenarios give an expected terminal 12-month Euribor of 4.2289%; the 95% range is 0.0000% to 25.0856%. Higher expected rates increase discounting of future positive cash flows and generally lower present values, all else equal.



Terminal Euribor distribution

Works Cited

- Bates, David S. "Jumps and Stochastic Volatility: Exchange Rate Processes Implicit in Deutsche Mark Options." *The Review of Financial Studies*, vol. 9, no. 1, 1996, pp. 69-107. <https://doi.org/10.1093/rfs/9.1.69>.
- Carr, Peter, and Dilip B. Madan. "Option Valuation Using the Fast Fourier Transform." *The Journal of Computational Finance*, vol. 2, no. 4, 1999, pp. 61-73.
- Cox, John C., Jonathan E. Ingersoll, Jr., and Stephen A. Ross. "A Theory of the Term Structure of Interest Rates." *Econometrica*, vol. 53, no. 2, 1985, pp. 385-407. <https://doi.org/10.2307/1911242>.
- Heston, Steven L. "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options." *The Review of Financial Studies*, vol. 6, no. 2, 1993, pp. 327-343. <https://doi.org/10.1093/rfs/6.2.327>.
- Lewis, Alan L. "A Simple Option Formula for General Jump-Diffusion and Other Exponential Levy Processes." SSRN, 2001, <https://ssrn.com/abstract=282110>.
- WorldQuant University. MScFE 622 Stochastic Modeling: Group Work Project 1. 2022. Course handout.